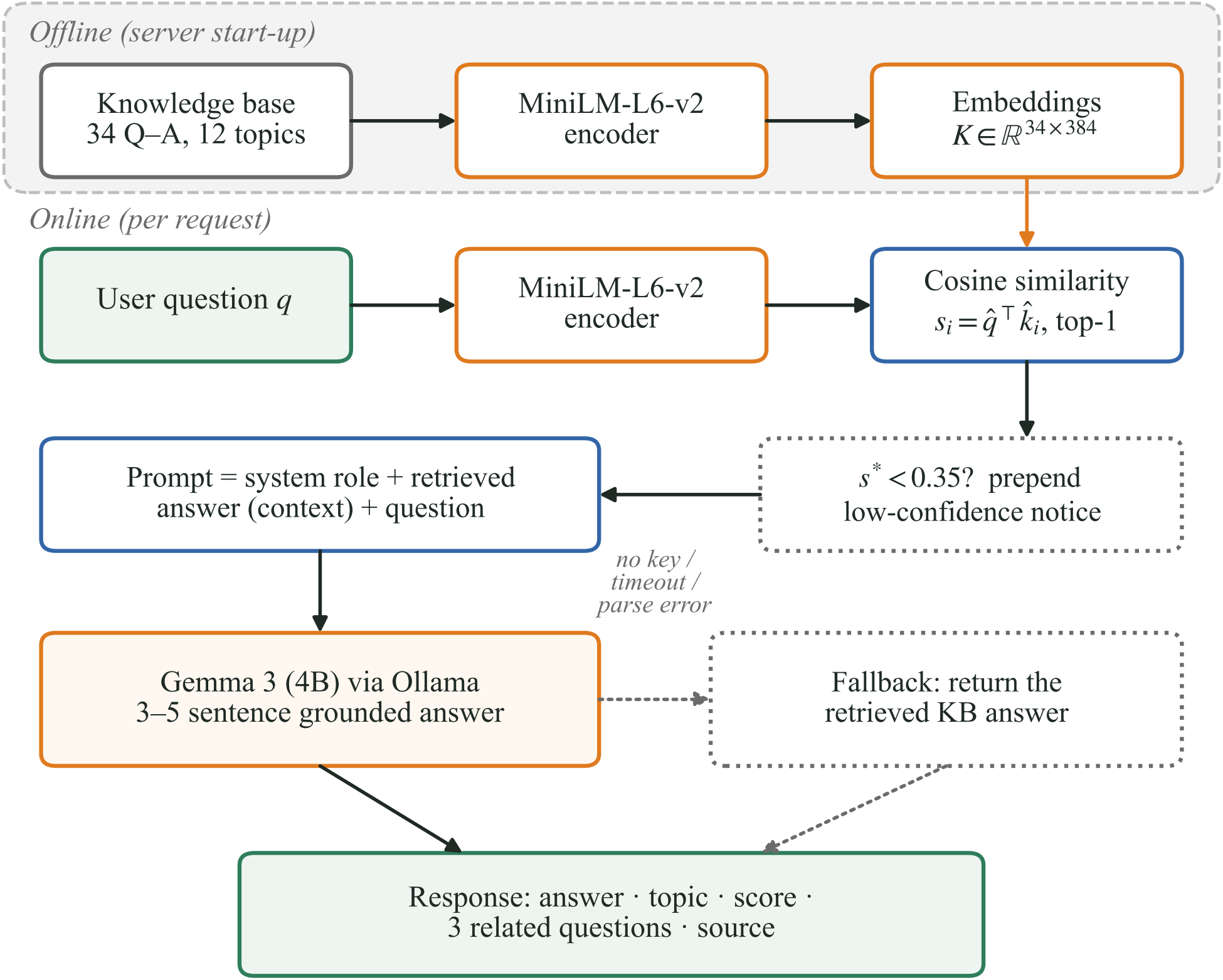
Offline (server start-up)
Knowledge base
34 Q–A, 12 topics
MiniLM-L6-v2
encoder
Embeddings
$K \in \mathbb{R}^{34 \times 384}$
Online (per request)
User question q
MiniLM-L6-v2
encoder
Cosine similarity
$s_i = \hat{q}^\top \hat{k}_i$, top-1
$s^* < 0.35$? prepend
low-confidence notice
Prompt = system role + retrieved
answer (context) + question
no key /
timeout /
parse error
Gemma 3 (4B) via Ollama
3–5 sentence grounded answer
Fallback: return the
retrieved KB answer
Response: answer · topic · score ·
3 related questions · source